INCH-POUND

MIL-F-80079C
6 August 1990
SUPERSEDING
22 February 1979
MIL-F-80079B

MILITARY SPECIFICATION

FURNACES, HEAT TREATING, ELECTRIC CONTROLLED ATMOSPHERE, HORIZONTAL BOX DESIGN

This specification is approved for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers heat-treating furnaces of the electrically heated, controlled atmosphere, horizontal box design.

1.2 Classification. The furnaces shall be of the following classes. The class to be furnished shall be as specified (see 6.2).

1.2.1 Classes.

Class 1 - 1400° F to 2000° F operating temperature range.
Class 2 - 1600° F to 2500° F operating temperature range.
Class 3 - 2000° F to 2750° F operating temperature range.

1.2.2 Sizes. The size of the furnace chamber shall be specified (see 6.2). The minimum size of the furnace chamber shall be not less than 6 inches in width by 12 inches in length by 6 inches in height. Size changes shall increase in increments of 3 inches in width or height and 6 inches in length up to a chamber size of 18 W by 30 L by 18 H. Chambers larger than 18 W by 30 L by 18 H shall increase in increments of 6 inches in any dimension as required. The inside dimensions of the heating chamber shall be not less than the dimensions specified. The door opening size shall be not less than width and height dimensions specified for the size furnace.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Industrial Plant Equipment Center, ATTN: DIPEC-SSM, 2163 Airways Boulevard, Memphis, Tennessee 38114-5051, by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3424
DISTRIBUTION STATEMENT A. Approved for public release, distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-3296	- Forges, Furnaces, and Ovens (exclusive of Space Heating and Cooking), Packaging of.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interferences.
MIL-STD-462	- Electromagnetic Interference Characteristics, Measurement of.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094).

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

U.S.DEPARTMENT OF LABOR, OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION (OSHA).

29 CFR 1910	- Occupational Safety and Health

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

Standards.

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

(Copies of specifications, standards, handbooks, drawings, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI C96.1 - Temperature Measurement Thermocouples.

(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Dept., 1430 Broadway, New York, NY 10018-3363.)

NATIONAL ELECTRICAL MANUFACTURERS' ASSOCIATION (NEMA)

NEMA ICS 1 - Industrial Controls and Systems, General Standards, for (DOD adopted)

(Application for copies should be addresed to the National Electrical Manufacturers' Association, 2101 L Street, N.W., Washington, DC 20037-1580.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA 79 - Electrical Standard for Industrial Machinery. (DOD adopted)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), one complete furnace shall be subjected to first article inspection (see 6.4) in accordance with 4.3.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

3.2 Design. The furnace shall be new (not a prototype) and one of the manufacturer's current production models capable of operations in accordance with the requirements herein. The furnaces shall be of the horizontal box design, having a hearth to support the charge and a single opening located in the end of the chamber for inserting and removing the charge in a horizontal plane. Unless otherwise specified, the furnaces shall employ the radiation method of heating with the charge directly exposed to the heat source. When required (see 6.2), the furnaces shall be provided with a retort to separate the heating elements from the atmosphere. Each furnace shall be of gas-tight construction and designed and equipped for operation with the type of controlled atmosphere specified (see 6.2). The furnaces shall be floor mounted and shall not require either excavation or a foundation.

3.2.1 Measuring and indicating device calibrations. Measuring and indicating devices shall be graduated in inch-pound (see 6.5.1), metric (see 6.5.2), or dual (inch-pound and metric) units as specified by the procuring activity (see 6.2). Regardless of the measurement system used, all measuring and indicating devices on the furnace shall be graduated in the same system.

3.2.2 Material. All material used in fabrication of the furnaces and related equipment shall be of the quality necessary to produce furnaces to meet the requirements described herein. Materials that are exposed to the internal environment of the furnace chamber shall be compatible with the type of atmosphere and temperature conditions specified for the furnace.

3.2.3 Reclaimed materials. The furnace may contain reclaimed materials to the maximum extent possible provided such materials will not jeopardize the intended use, performance, or design life of the machine. Reclaimed materials shall have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used, rebuilt, or remanufactured end products are allowed under this specification.

3.2.4 Energy efficiency. The furnace and its components that consume energy in normal operation shall be designed and constructed for energy efficiency in accordance with the developments available within the industry.

3.2.5 Controls. All mechanical, hydraulic, and pneumatic operating controls shall be located convenient to the operator's work station(s).

3.2.6 Safety and health requirements. All parts, components, mechanisms, and assemblies furnished on the furnace, whether or not specifically required herein, shall conform to all requirements of OSHA 29 CFR 1910. Additional safety and health requirements shall be as specified (see 6.2). Covers, guards, or other safety devices normally furnished as standard on the manufacturer's commercial furnace supplied to the commercial market shall be provided for the point of operation and all other parts of the furnace that present safety hazards. If point-of-operation guarding is influenced by workpiece size and configuration, the procuring activity shall provide part data to the supplier of the machine (see 6.2). The supplier of the machine shall provide the special guarding which shall meet OSHA 29 CFR 1910.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

3.2.7 Mercury restriction. The furnace shall not contain mercury or mercury compounds nor be exposed to free mercury during manufacture.

3.2.8 Asbestos restriction. Asbestos and materials containing asbestos shall not be used on or in the furnace.

3.2.9 Environmental protection. During the manufacture, operation, service, transportation, or storage of the machine and its components, materials hazardous to the ecological system as prohibited by applicable Federal, state, or local statutes in effect in effect on the date the contract shall not be used or emitted.

3.2.10 Lubrication. When necessary to reduce wear and maintain efficiency, means shall be provided for lubrication of all sliding and rotating parts, except nonadditive pack-seal bearings. Lubrication systems shall be designed and built to prevent any contamination of atmosphere, components, material or equipment. Lubrication points shall be readily accessible, and identified in the operator's manual with type of lubricant and instructions for application.

3.2.11 Interchangeability. To provide for replacement of worn parts, all parts shall be manufactured to definite dimensions and tolerances that will permit installation of replacement parts without modification of the part or machine. Replacement parts shall be domestically available.

3.3 Construction. The furnace shall be constructed of parts which are new, without defects, and free of repairs. The structure shall withstand all specified power encountered during operation of the furnace to its maximum rating and capacity without permanent distortion or failure.

3.3.1 Castings and forgings. All castings and forgings shall be free of defects, scale, and mismatching. No processes such as welding, peening, plugging, or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing, or soldering. Welding, brazing, or soldering shall be employed only where specified in the original design. Any material used for such operations shall be thoroughly removed from the part(s) upon completion of the operations.

3.3.3 Fastening devices. All fasteners shall be installed to prevent change of tightness. Fastening devices subject to removal or adjustment shall not be permanently installed.

3.3.4 Surfaces. All surfaces shall be clean and free of harmful or extraneous materials. All edges shall be either rounded or beveled unless sharpness is required to perform a necessary function. Except as otherwise specified herein, the condition and finish of all surfaces shall be in accordance with the manufacturer's commercial practice.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.3.5 Painting. Unless otherwise specified (see 6.2), the paint color and application shall be in accordance with the manufacturer's commercial practice.

3.3.6 Threads. All threaded parts used on the furnace and its related attachments and accessories shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.7 Electromagnetic interference control. When specified (see 6.2), the equipment furnished under this specification shall comply with MIL-STD-461. The equipment and subsystem(s) class and the emission and susceptibility requirements shall be as specified.

3.4 Components. The furnaces shall consist of not less than a frame, shell, hearth, refractory, door, flame curtain, heating elements, thermocouples, and instruments.

3.4.1 Frame. The furnace shall have a structural steel frame, and shall provide a stable support for the work chamber and a maximum load. If legs are used, they shall be braced and provided with bearing plates. The working height of the furnace hearth shall be between 36 and 42 inches above floor level.

3.4.2 Shell. The outer shell of the furnace shall be constructed of steel plate, not less than 3/16 inch in thickness. The shell shall be reinforced with steel members, and securely welded at all joints, except openings required for repair and rebuild shall be provided with suitable gaskets and bolted. The shell shall be of gas-tight construction, and provided with suitable threaded connections for operation with a controlled atmosphere of the type specified for each furnace. The front of the furnace surrounding the door opening shall be designed and constructed of such material that warpage will be prevented.

3.4.3 Hearth. The hearth shall be designed to permit maximum heat transfer and uniform temperature throughout the work chamber. Furnaces with maximum operating temperatures above 2000^{0} F shall have hearth plates constructed of silicon carbide material. Each furnace hearth shall be capable of supporting a load of not less than 100 pounds of steel per square foot of hearth area without visible deflection. The hearth plates shall be easily removed and replaced with means provided to allow unrestricted movement for expansion and contraction. When specified (see 6.2), the furnace hearth shall be provided with slide or roller rails for moving the charge in and out of the work chamber. When required, the chamber shall be provided with open-grid type shelves. Number and arrangement of shelves shall be as specified (see 6.2).

3.4.4 Refractory. The lining shall consist of suitable refractory or insulating brick, selected to minimize atmosphere contamination, resist abrasion and spalling and backed up with insulated brick, block or preformed insulation. Overall thickness and quality of the refractory and insulation shall be such that the outside surface temperature of the Class I furnace will not exceed 100^{0} F, or the Class 2 furnace 150^{0} F, or the Class 3 furnace

Source: https://assist.dla.mil - Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

200° F above ambient air after the temperature has stabilized at maximum operating temperature. When motar is used, it should be of the same basic material as the units being cemented together. Powdered, shredded or granular insulation shall not be used for insulating purposes.

3.4.5 Door. Unless otherwise specified, the door shall be of the vertical lift type, and manually operated with suitable counter weights. When required (see 6.2), the door shall be pneumatically operated with a pressure regulator, cutoff valve and a foot control valve furnished. When operating on an air pressure of 80 PSI, the door shall fully open or close in not more than 10 seconds. The door shall be fully insulated with refractory or insulating material, and shall overlap the brick facing of the furnace not less than 2 and 1/2 inches on all sides. The door shall be provided with jambs and sufficient seals to form a gas-tight unit when closed. The door shall be provided with a nonfogging sight glass for observing the charge without opening the door.

3.4.6 Flame curtain. Unless otherwise specified (see 6.2), a gas-air-flame curtain shall be provided for the door openings, and shall operate automatically by limit switches through solenoid valves to protect the furnace atmosphere when the door is open.

3.4.7 Heating elements. Heating elements shall be distributed and located to meet the heating requirements and temperature uniformity specified herein. The elements shall be designed and mounted in accordance with the best accepted commercial practice for the particular type of atmosphere and temperature range specified for each furnace. Heating elements shall be fabricated from material that is compatible with the atmosphere and temperature specified for each furnace and shall be capable of continuous operation at any temperature within the operating range of the furnace. Furnaces equipped with silicon carbide or similar elements that require voltage compensation for change in resistance shall be provided with a multiple-tap transformer having not less than seven steps. An ammeter or a comparable instrument shall be furnished with furnaces having silicon carbide or equivalent type elements to detect changes in resistance. The power density on the heating elements shall not exceed 15 watts per square inch of radiation surface for alloy type elements, and 50 watts for silicon carbide or equivalent elements. When specified (see 6.2), the furnace shall have radiant tubes of sufficient number to provide a uniformly heated furnace chamber, eliminating the need for a four-sided retort.

3.4.8 Thermocouples. Furnaces having a temperature rating in excess of 2000° F shall be provided with platinum-rhodium type thermocouples. Furnaces having a temperature rating of 2000° F and lower shall be provided with chromel-alumel type thermocouples. The thermocouples shall be adequately protected from contamination of furnace atmosphere by means of suitable protective tubes. The thermocouples shall conform to ANSI C96.1. Thermocouple lead wires shall be not less than 25 feet in length and of the same material as the thermocouples, or of a material having the same thermoelectrical characteristics at a thermocouple head temperature between 0 degrees and 250° F.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

3.4.9 Program control. When required, the furnace shall be provided with a program control system integrated with the instrument controls. Programing requirements such as heating rates, cooling rates, and soak periods shall be as specified (see 6.2).

3.4.10 Instruments. Each furnace shall be provided with a single, self-contained temperature indicating-recording-controller type instrument with an accuracy of 0.25 percent or better of full scale reading. Solid-state design shall be employed where possible. The instrument shall be marked to indicate the particular type of thermocouple for which it is calibrated. Automatic reference junction compensation shall be provided in all instruments using thermocouples as the primary temperature sensing element.

3.4.10.1 Temperature recorder. The temperature recorder shall be of the strip chart, self-balancing type. Unless otherwise specified, the calibrated chart width shall be not less than 6 inches, and the chart length shall be not less than 80 feet to the roll. The chart speed shall be 1 inch per hour. The chart mechanism shall be furnished in a dust-proof housing with a glass door that permits full view of chart width, and not less than 7 inches of chart length. When required, the chart width shall not be less than 9 inches and provisions provided for changing chart speed to 4 and 6 inches per hour as specified (see 6.2).

3.4.10.2 Temperature controller. Unless otherwise specified, the temperature controller shall be of the time proportioning type control, and shall operate a contactor through an intermediate relay system. When required, the controller shall be of the current proportioning type control, and shall operate either a saturable core reactor or silicon controlled rectifier as specified (see 6.2). Either system furnished shall have the necessary adjustments to be capable of maintaining the set control temperature within plus 2° F. The control system shall provide thermocouple upscale failsafe protection that will shut off the heat source when the chamber temperature exceeds approximately 100° F above maximum.

3.4.10.3 Temperature indicator. The temperature indicator scale shall be approximately the same length as the temperature recorder chart width. The calibrated scale range shall not be less than 100° F above the maximum temperature of the furnace. The increments between successive temperature marks shall be not greater than 20° F, with the spacing of numbered marks not greater than 400° F. The control indicator shall be provided with two pointers. One pointer shall be adjustable to the desired control temperature, and one pointer shall indicate the actual furnace temperature.

3.4.10.4 Excess temperature control. The furnace shall be provided with an excess temperature controller. The excess temperature control circuit shall be independent of the operating control circuitry. The controller shall provide for selecting the over-temperature control point, and shall function to shut off the heat source to the furnace when the over-temperature control point is reached. In case the furnace is shut down by the excess temperature control system, the system shall sound an audible alarm, and the furnace shall remain off until started manually. Calibrated accuracy shall be within one percent of full scale reading.

Source: https://assist.dla.mil Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

3.4.10.5 Instrument panel. Floor mounted instrument panels shall be provided for mounting all instruments and accessory controls. All ventilated openings shall be designed to prevent entrance of any deleterious substance. Signal panel lights shall be provided to indicate the furnace cycle performance. The panels shall provide flush or semi-flush mounting of all instruments.

3.4.10.6 Thermocouple burn-out alarm signal. The excess temperature control and temperature controller shall be designed and equipped to provide an audible and visible alarm signal in the event of the loss of either or both thermocouple circuits.

3.4.11 Electrical equipment. The furnaces shall be furnished with electrical motors, contactors, controls, relays, transformers, signal lights, and switches, pre-wired where possible for operation upon connection to the source of power. A fusible disconnect switch or circuit breaker conforming to NEMA ICS 1 requirements, shall be provided to disconnect all power to the furnace. The furnace shall be initally wired for operation as specified (see 6.2).

3.4.11.1 Power supply. The furnaces shall operate from a single source primary power supply. A stepdown transformer shall be provided for the 120 volt control system power supply. The furnace shall be designed for operating from the power supply specified (see 6.2).

3.4.11.2 Electrical connections. Electrical connections of conductors and terminal parts shall be of the pressure or solder type. When soldered connections are used, the conductor and terminal parts shall be mechanically secured before soldering. Flux for soldering shall be of a resin base.

3.5 Size and capacity. Unless otherwise specified (see 6.2), the size and capacity of the furnace shall meet the requirements specified in 1.2.1.

3.6 Performance. Each furnace shall be capable of continuous operation at any temperature within its operating range, with loads up to 100 lbs.per sq.foot of hearth area.

3.6.1 Heating rate. The Class 1 furnace shall be capable of heating a charge of low carbon steel equivalent to not less than 40 pounds per square foot of hearth area from 70° F to maximum temperature in one hour. Class 2 and Class 3 furnaces shall be capable of heating an equivalent charge to 2200° F in 1 hour. The charge is based on gross load and includes the weight of baskets, trays, fixtures, required to retain the workload. The heating rate is determined by loading the charge in the chamber which has been stabilized at the specified temperature.

3.6.2 Temperature uniformity. With the furnace charged and the temperature stabilized, the temperature at any point in the effective working zone shall not vary more that plus or minus 25° F from the instrument control

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

point. Each furnace shall be capable of maintaining the temperature uniformity specified with any size charge up to maximum load and at any desired temperature within its operating range.

3.7 Standard equipment. All standard equipment normally provided with the manufacturer's commercial machine shall be furnished.

3.8 Optional equipment. Optional equipment shall be furnished as specified and fully described (see 6.2). The optional equipment provided shall be functional without requiring modification of the equipment or machine.

3.9 Marking on instruments, control panels, charts, and plates. All words on instruments, control panels, charts, and plates shall be in the English language. Characters shall be permanently marked in boldface on a contrasting background. All plates shall be corrosion resistant.

3.10 Nameplate. A nameplate shall be securely attached to each machine. Unless otherwise specified (see 6.2), the nameplate shall contain the information listed below.

Nomenclature
Manufacturer's name
Manufacturer's model designation
Manufacturer's serial number
Power input (volts, total amps, phase, frequency)
Amp rating of largest motor
Short circuit/over-current protection rating
Contract number or order number
National Stock Number or Plant Equipment Code
Date of manufacture

3.11 Technical data. When technical data is required it shall be furnished in accordance with the requirements of the contract. All technical data furnished shall be written in the English language.

3.12 Workmanship. Workmanship of the furnace and its accessories shall meet all requirements specified herein.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspections set forth in this specification shall become a part of the contractors overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submissions of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4).

b. Quality conformance inspection (see 4.5).

c. Acceptance test (see 4.6).

4.3 Inspection conditions. Unless otherwise specified (see 6.2), all inspections shall be performed in accordance with the test conditions specified in 4.8.1.

4.4 First article inspection. When a first article inspection is required, it shall be applied to the first article submitted in accordance with 3.1. Unless otherwise specified (see 6.2), first article inspection shall consist of the examination in 4.7 and all tests in 4.8. The furnace shall pass the examination and all tests to be accepted.

4.5 Quality conformance inspection. Quality conformance inspection shall be applied to each furnace prior to being offered for acceptance under the contract. Unless otherwise specified (see 6.2), quality conformance inspection shall consist of the examination in 4.7, the tests in 4.8, and the inspection in 4.9. The machine shall pass the examination, all tests, and the inspection to be accepted.

4.6 Acceptance test. Acceptance test shall be performed on each furnace to ensure conformance with this specification. Unless otherwise specified (see 6.2), the acceptance test shall be performed only after the furnace is installed at its final location. The acceptance test shall consist of the examination in 4.7 and all tests in 4.8. The machine shall pass the examination and all tests to be accepted.

4.7 Examination. Each furnace shall be examined to determine compliance with all requirements of this specification.

4.8 Tests.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

4.8.1 Test conditions. All tests shall be performed in an indoor facility with ambient conditions of 41° F to 104° F and 20 to 95 percent relative humidity.

4.8.2 Instrument test. Indicating and recording instruments shall be tested by impressing various predetermined voltages, simulating thermocouple outputs for different furnace temperatures to terminals where normally the thermocouples would be connected. These tests shall be made in accordance with, and results checked against, standards established in ANSI C96.1. The instruments shall be tested through their operating range.

4.8.3 Circuit test. Each circuit of the electrical system shall be tested for dielectric strength, continuity, insulation-resistance and faulty grounds in accordance with the requirements of NEMA ICS 1.

4.8.4 Electromagnetic interference control test. Equipment requiring electromagnetic interference control shall be tested for compliance with 3.4.4 using the procedures given in MIL-STD-462.

4.8.5 Operational test. The furnace shall be prepared for operation, and operated in accordance with the manufacturer's handbook of instructions until the refractory and insulation has cured. With the chamber temperature stabilized at 2000° F for the Class 1 furnace and at 2200° F for the Class 2 and Class 3 furnaces, the chamber shall be loaded with a charge of iron or steel equivalent to not less than 40 pound per square foot of hearth area. At the end of one hour, the chamber temperature shall be within 10° F of the initial temperature. Each safety device shall be tested by simulating a failure that normally would cause the device or system to operate. Each system shall perform its intended function.

4.8.6 Performance test. The furnace shall be charged as specified in 4.8.5, and a temperature survey performed in the effective work zone of the heating chamber. The effective work zone (test zone) is defined as being 1/6 less than the actual cubic volume of chamber air space. A minimum of nine test locations shall be used, with one thermocouple located in each corner of the test zone, and one thermocouple located within three inches of the control thermocouple. For furnaces with a chamber volume of less than 10 cubic feet, the test may be performed with a minimum of three thermocouples, with one each located at the front, center and rear of the test zone. The survey shall be performed at the minimum and maximum temperature range of Class 1 furnace, and the minimum and at 2200° F of the Class 2 and Class 3 furnaces. From the time the heating cycle is initiated until the thermal equilibrium is reached, none of the temperature readings should exceed the maximum of the temperature range being surveyed. After thermal equilibrium is reached, readings shall be taken at 5 minute intervals for 30 minutes. The maximum temperature variation between all points under test shall not vary more than plus or minus 25° F from the instrument control point.

4.8.7 Chamber surface temperature test. The chamber temperature shall be stabilized at maximum operating temperature, and the surface temperature of the chamber and door checked at random locations. The surface temperature shall not exceed the limits specified in 3.4.4, except within 3 inches of

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:32Z
Check the source to verify that this is the current version before use.

through-fastening devices. The Class 2 and Class 3 furnace shall be unloaded before increasing the temperature above the melting point of the charge.

4.9 Packaging inspection. Packaging of each item shall be inspected to determine compliance with the requirements of section 5.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-F-3296.

6. NOTES

6.1 Intended use. Furnaces covered by this specification are intended for use in heat treating shops, in such processes as hardening, tempering, normalizing and annealing metal parts, utilizing a controlled manufactured atmosphere.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Class and size required (see 1.2.1).

c. Size required (see 1.2.2).

d. Issue of DODISS to be cited in the solicitation and, if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).

e. First article, if required (see 3.1).

f. Retort, if required (see 3.2).

g. Type of atmosphere the furnace shall be designed and equipped to use, such as endothermic, exothermic, or ammonia gas, (see 3.2).

h. System of units for measuring and indicating devices (see 3.2.1).

i. Additional safety and health requirements, if required (see 3.2.6).

j. Specify configuration of additional point-of-operation guarding, if required (see 3.2.6).

k. Painting, if different (see 3.3.5).

l. If electromagnetic interference control is required, specify the equipment and subsystem class and the emission and susceptibility required (see 3.3.7).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

m. Hearth slide or roller rails, when required (see 3.4.3).

n. Chamber shelves, specify number and spacing when required (see 3.4.3).

o. Power operated door, when required (see 3.4.5).

p. Flame curtain, if different (see 3.4.6).

q. Radiant Tubes, when required (see 3.4.7).

r. Program control, and programing requirements when required (see 3.4.9).

s. Temperature recorder chart width and speed, if different (see 3.4.10.1).

t. Temperature controller, specify saturable core reactor or silicon controlled rectifier, if required (see 3.4.10.2).

u. Initial voltage, if different (see 3.4.11).

v. Power supply, if different (see 3.4.11.1).

w. Size and capacity, if different (see 3.5.).

x. Optional equipment, when required (see 3.8).

y. Nameplate,if different (see 3.10).

z. Inspection conditions, if different (see 4.3).

aa. First article inspection, if different (see 4.4).

bb. Quality conformance inspection, if different (see 4.5).

cc. Acceptance test, if different (see 4.6).

6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DIDs) should be reviewed in conjunction with the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
N/A	DI-TMSS-80527	Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English Language

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

N/A	DI-TMSS-80528	Supplemental Data for Commercial Off-The-Shelf (COTS) Manuals	Require all data be in the English language

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.3.1 Technical data. All technical data ordered using a DD Form 1423 should be written in the English language.

6.4 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.5 Definitions.

6.5.1 Inch-pound units. Inch-pound units are a system of measures based on the yard and pound commonly used in the United States of America and defined by the National Institute of Standards and Technology. Inch-pound units having the same names in other countries may differ in magnitude. 6.5.2

6.5.2 Metric units. Metric units are a system of basic measures defined by the International System of Units based on "Le Systeme International D'Unites (SI)," of the International Bureau of Weights and Measures. These units are described in ASTM E 380 and IEEE 268 (MIL-STD-961).

6.6 Measurement System. In this specification, all measurements, dimensions, sizes, and capacities are given in inch-pound units. These measurements may be converted to metric units through the use of the conversion factors and methods specified in FED-STD-376.

6.7 Changes from previous issues. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

6.8 Safety and health requirements. Standard 29 CFR 1910 limits only the total hazard level (noise, radiation, electromagnetic emissions, noxious vapors, air contaminants, and heat) of the environment in which a machine will

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

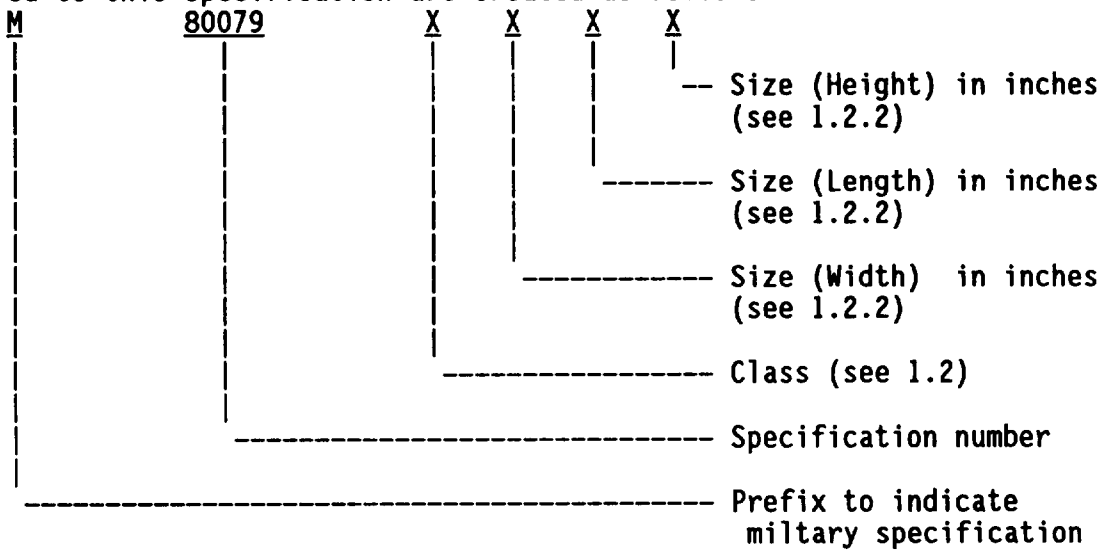
operate, It does not limit the hazard level of individual machines in an operating environment.

6.9 Training. Training required by the procuring activity should be provided as specified in the contract.

6.10 Warranty. Warranty requirements should be as specified by the procuring activity in the contract.

6.11 Inspection location. The contractor should identify in his response to the solicitation the location where inspection and tests are to be performed.

6.12 Part or Identifying Number (PIN). The PIN to be used for machines acquired to this specification are created as follows:



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

6.13 Subject term (keyword) listing.

Flame curtain
Hearth
Horizontal box
Program control

Custodians:
Army - AL
Navy - SH
Air Force - 99
Review activities:
Navy - AS
Air Force - 84
DLA - GS

User activities:
Army - AV
Navy - MC, OS

Preparing activity:
DLA - IP

Project (3424-0116)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-F-80079C	2. DOCUMENT DATE (YYMMDD) 900806

3. DOCUMENT TITLE
Furnaces, Heat Treating, Electric Controlled Atmosphere, Horizontal Box Design

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Defense Industrial Plant Equipment Center	b. TELEPHONE (Include Area Code) (1) Commercial: 901-775-6257 (2) AUTOVON: 683-6257
c. ADDRESS (Include Zip Code) 2163 Airways Blvd. ATTN: DIPEC-SSM, Memphis, TN 38114-5051	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:32Z
Check the source to verify that this is the current version before use.